



KENYATTA UNIVERSITY

**EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN
ENGINEERING**

EEE 201: THERMODYNAMICS

2ND SEMESTER 2020/2021

TIME: 2 HRS

- This Paper consists of FIVE QUESTIONS
- Answer Question ONE any Other TWO.
- Steam Tables are provided

QUESTION ONE (COMPULSORY) 30 MARKS

- (a) Define the following terms used in Engineering Thermodynamics giving examples.
- (i) Intensive property
 - (ii) Open system
 - (iii) Closed system
 - (iv) Isolated system **(4 marks)**
- (b) Explain the fundamental differences between the First Law of thermodynamics and Second law of Thermodynamics **(2 marks)**
- (c) State Kelvin Plank and Clausius statement of the Second Law of Thermodynamics **(2 Marks)**
- (d) Write the equation for the displacement of work for the following processes with a PV diagram:
- (i) Constant volume process
 - (ii) Constant pressure process
 - (iii) Isothermal process
 - (iv) Polytropic process **(4 Marks)**

(e) A fluid undergoes a reversible adiabatic compression process from 0.5MPa, 0.2m³ to 0.05m³ according to the law $PV^{1.3}=C$. Determine the following during the process:

- (i) Change in enthalpy
- (ii) Change in internal energy
- (iii) Heat transfer
- (iv) Work transfer (6 Marks)

(f) (i) State and explain Fourier's law of conduction (2 Marks)

(ii) Explain two mechanisms through which heat transfer by conduction takes place in solids (1 Marks)

(i) Explain the analogy between heat transfer by conduction and flow of electricity through ohmic resistance. (2 Marks)

(g) A cold storage room has walls made of 200 mm of brick on the outside, 80 mm of plastic foam, and finally 20 mm of wood on the inside. The outside and inside air temperatures are 25°C and -3°C respectively. If the outside and inside convective heat transfer coefficients are respectively 10 and 30 W/m²°C, and the thermal conductivities of brick, foam and wood are 1.0, 0.02 and 0.17 W/m°C respectively. Determine:

(i) Overall heat transfer coefficient (3 Marks)

(ii) The rate of heat removed by refrigeration if the total wall area is 100m² (2 Marks)

(iii) Outside and inside surface temperatures and mid-plane temperatures of composite wall. (2 Marks)

QUESTION 2 (20 MARKS)

(a) Show that for a perfect gas undergoing Polytropic expansion or compression,

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{n-1}{n}} = \left(\frac{v_1}{v_2}\right)^{n-1} \text{ where } T, P, v \text{ and } \gamma \text{ are Temperature, pressure, volume and adiabatic index respectively.}$$

(4 Marks)

(b) The following cycle involves 3 kg of air : Polytropic compression from state 1 to 2 where pressure $P_1 = 150$ kPa, temperature $T_1 = 360$ K, pressure $P_2 = 750$ kPa and Polytropic index

$n = 1.1$; isobaric cooling from state 2 to 3; and finally isothermal heating from state 3 to 1.

- i) Draw the P - V diagram for the cycle (1 marks)
- ii) Find temperatures, pressures and volumes at each state (5 marks)
- iii) Determine the net work (2 marks)
- iv) Calculate the net heat absorbed or rejected. (2 Marks)

(c) A quantity of air occupies a volume of 30 litres at a temperature of 38°C and pressure 104 kN/m^2 . The temperature of the air can be raised;

- i) by isochoric heating process until the pressure is 208 kN/m^2
- ii) by adiabatic compression until volume is 6 litres.

Find for each case the final temperature, the external work done, the change in internal energy and the heat transferred. Take $R = 0.29 \text{ kJ/kg K}$ and $\gamma = 1.4$ (6 Marks)

QUESTION 3 (20 MARKS)

(a) Starting from the First law of Thermodynamics, derive steady flow energy equation for a stream of fluid entering and leaving a control volume (4 Marks)

(b) The working fluid flows at a rate of 220 kg/min in a steady flow process. The system rejects 100 kJ/s of heat passing through the system. The condition of the fluid at inlet and outlet are: $C_1 = 220 \text{ m/s}$, $p_1 = 6.0 \text{ bar}$, $u_1 = 2000 \text{ kJ/kg}$, $v_1 = 0.36 \text{ m}^3/\text{kg}$ and $C_2 = 140 \text{ m/s}$, $u_2 = 1400 \text{ kJ/kg}$ and specific volume $v_2 = 1.3 \text{ m}^3/\text{kg}$, where the suffix 1 and 2 indicate the inlet and outlet conditions respectively. Neglecting changes in potential energy Determine:

- (i) The power output from the system in MW (5 Marks)
- (ii) Inlet and outlet pipe diameter used to convey the steam (3 Marks)

- (c) A flow-rate of 12kg/min of a fluid goes through a reversible steady flow process. The properties of the fluid at inlet are $P_1 = 1.4\text{bar}$, $v_1 = 0.04\text{m}^3/\text{kg}$, $C_1 = 120\text{m/s}$ and $u_1 = 120\text{kJ/kg}$. At exit $P_2 = 5.6\text{bar}$, $v_2 = 0.2\text{m}^3/\text{kg}$, $C_2 = 180\text{m/s}$ and $u_2 = 720\text{kJ/kg}$. During passage, the fluid rejects 60kJ/s of heat and rises through an elevation of 60m.

Determine:

- (j) The power output from the system (5 Marks)
(ii) Inlet and outlet pipe cross-sectional areas (3 Marks)

QUESTION 4 (20 MARKS)

- (a) Define dryness fraction of steam (1 Mark)
(b) Using a temperature-enthalpy diagram, show the different phases of steam generation indicating type of enthalpy at each phase. (4 Marks)
(c) Steam at a pressure of 2.4bar and with dryness fraction 0.85 is compressed adiabatically to a pressure of 6.5bar. The law of compression is $PV^{1.13} = C$. Determine:
i) The final dryness fraction of the steam
ii) The final density of the steam
iii) The change in internal energy of the steam (6 Marks)
(d) Steam is supplied superheated at 42bar to a turbine and the condenser pressure is 0.035 bar. If the plant operates on the Rankine cycle, calculate, per kg of steam:
i. The work output neglecting the feed pump work;
ii. The heat supplied;
iii. The specific steam consumption
iv. Thermal Efficiency of the plant
v. (9 Marks)

QUESTION 5 (20 MARKS)

- (a) Show that the air standard thermal efficiency for an OTTO cycle is given by

$$1 - \frac{1}{r^{\gamma-1}} \quad (6 \text{ marks})$$

Where r = Compression ratio and γ = Adiabatic index

(b) In a standard OTTO cycle, the maximum and the minimum temperatures are 1400°C and 15°C . The heat supplied per kg of air 800kJ . Calculate

(i) The compression ratio;

(ii) Maximum and minimum pressures in the cycle

(iii) Thermal efficiency

(6 Marks)

(c) A gas turbine has an overall pressure ratio of $9/1$ and a maximum cycle temperature of 650°C . The turbine drives the compressor and an electric generator. The ambient temperature is 15°C and the isentropic efficiencies of the compressor and turbine are 0.8 and 0.84 respectively. Calculate the power output in kilowatts for an air flow of 20kg/s . Calculate also the thermal efficiency and the work ratio. Neglect changes in kinetic energy, and the loss of pressure in the combustion chamber. Take for air $C_p = 1.005\text{kJ/kgK}$ and $\gamma = 1.4$. For combustion gases $C_p = 1.15\text{kJ/kgK}$ and $\gamma = 1.33$

(8 Marks)

T [°C]	p_s [bar]	v_g [m ³ /kg]	h_f [kJ/kg]	h_g [kJ/kg]	s_f [kJ/kg K]	s_{fg} [kJ/kg K]	s_g [kJ/kg K]
0.01	0.006112	206.1	0*	2500.8	2500.8	0†	9.155
1	0.006566	192.6	4.2	2498.3	2502.5	0.015	9.113
2	0.007054	179.9	8.4	2495.9	2504.3	0.031	9.071
3	0.007575	168.2	12.6	2493.6	2506.2	0.046	9.030
4	0.008129	157.3	16.8	2491.3	2508.1	0.061	8.989
5	0.008719	147.1	21.0	2488.9	2509.9	0.076	8.948
6	0.009346	137.8	25.2	2486.6	2511.8	0.091	8.908
7	0.01001	129.1	29.4	2484.3	2513.7	0.106	8.868
8	0.01072	121.0	33.6	2481.9	2515.5	0.121	8.828
9	0.01147	113.4	37.8	2479.6	2517.4	0.136	8.788
10	0.01227	106.4	42.0	2477.2	2519.2	0.151	8.749
11	0.01312	99.90	46.2	2474.9	2521.1	0.166	8.710
12	0.01401	93.83	50.4	2472.5	2522.9	0.180	8.671
13	0.01497	88.17	54.6	2470.2	2524.8	0.195	8.633
14	0.01597	82.89	58.8	2467.8	2526.6	0.210	8.594
15	0.01704	77.97	62.9	2465.5	2528.4	0.224	8.556
16	0.01817	73.38	67.1	2463.1	2530.2	0.239	8.518
17	0.01936	69.09	71.3	2460.8	2532.1	0.253	8.481
18	0.02063	65.08	75.5	2458.4	2533.9	0.268	8.444
19	0.02196	61.34	79.7	2456.0	2535.7	0.282	8.407
20	0.02337	57.84	83.9	2453.7	2537.6	0.296	8.370
21	0.02486	54.56	88.0	2451.4	2539.4	0.310	8.334
22	0.02642	51.49	92.2	2449.0	2541.2	0.325	8.297
23	0.02808	48.62	96.4	2446.6	2543.0	0.339	8.261
24	0.02982	45.92	100.6	2444.2	2544.8	0.353	8.226
25	0.03166	43.40	104.8	2441.8	2546.6	0.367	8.190
26	0.03360	41.03	108.9	2439.5	2548.4	0.381	8.155
27	0.03564	38.81	113.1	2437.2	2550.3	0.395	8.120
28	0.03778	36.73	117.3	2434.8	2552.1	0.409	8.085
29	0.04004	34.77	121.5	2432.4	2553.9	0.423	8.050
30	0.04242	32.93	125.7	2430.0	2555.7	0.436	8.016
32	0.04754	29.57	134.0	2425.3	2559.3	0.464	7.948
34	0.05318	26.60	142.4	2420.5	2562.9	0.491	7.881
36	0.05940	23.97	150.7	2415.8	2566.5	0.518	7.814
38	0.06624	21.63	159.1	2411.0	2570.1	0.545	7.749
40	0.07375	19.55	167.5	2406.2	2573.7	0.572	7.684
42	0.08198	17.69	175.8	2401.4	2577.2	0.599	7.620
44	0.09100	16.03	184.2	2396.6	2580.8	0.625	7.557
46	0.1009	14.56	192.5	2391.8	2584.3	0.651	7.494
48	0.1116	13.23	200.9	2387.0	2587.9	0.678	7.433
50	0.1233	12.04	209.3	2382.1	2591.4	0.704	7.371
55	0.1574	9.578	230.2	2370.1	2600.3	0.768	7.223
60	0.1992	7.678	251.1	2357.9	2609.0	0.831	7.078
65	0.2501	6.201	272.0	2345.7	2617.7	0.893	6.937
70	0.3116	5.045	293.0	2333.3	2626.3	0.955	6.800
75	0.3855	4.133	313.9	2320.8	2634.7	1.015	6.666
80	0.4736	3.408	334.9	2308.3	2643.2	1.075	6.536
85	0.5780	2.828	355.9	2295.6	2651.5	1.134	6.410
90	0.7011	2.361	376.9	2282.8	2659.7	1.192	6.286
95	0.8453	1.982	398.0	2269.8	2667.8	1.250	6.166
100	1.01325	1.673	419.1	2256.7	2675.8	1.307	6.048

† u and s are chosen to be zero for saturated liquid at the triple point.

Note: values of v_f can be found on p. 10.

p [bar]	T_s [°C]	v_R [m³/kg]	u_f [kJ/kg]	u_g [kJ/kg]	h_f [kJ/kg]	h_{fg} [kJ/kg]	h_g [kJ/kg]	s_f [kJ/kg K]	s_{fg} [kJ/kg K]	s_g [kJ/kg K]
0.006112	0.01	206.1	0†	2375	0*	2501	2501	0†	9.155	9.155
0.010	7.0	129.2	29	2385	29	2485	2514	0.106	8.868	8.974
0.015	13.0	87.98	55	2393	55	2470	2525	0.196	8.631	8.827
0.020	17.5	67.01	73	2399	73	2460	2533	0.261	8.462	8.723
0.025	21.1	54.26	88	2403	88	2451	2539	0.312	8.330	8.642
0.030	24.1	45.67	101	2408	101	2444	2545	0.354	8.222	8.576
0.035	26.7	39.48	112	2412	112	2438	2550	0.391	8.130	8.521
0.040	29.0	34.80	121	2415	121	2433	2554	0.422	8.051	8.473
0.045	31.0	31.14	130	2418	130	2428	2558	0.451	7.980	8.431
0.050	32.9	28.20	138	2420	138	2423	2561	0.476	7.918	8.394
0.055	34.6	25.77	145	2422	145	2419	2564	0.500	7.860	8.360
0.060	36.2	23.74	152	2425	152	2415	2567	0.521	7.808	8.329
0.065	37.7	22.02	158	2427	158	2412	2570	0.541	7.760	8.301
0.070	39.0	20.53	163	2428	163	2409	2572	0.559	7.715	8.274
0.075	40.3	19.24	169	2430	169	2405	2574	0.576	7.674	8.250
0.080	41.5	18.10	174	2432	174	2402	2576	0.593	7.634	8.227
0.085	42.7	17.10	179	2434	179	2400	2579	0.608	7.598	8.206
0.090	43.8	16.20	183	2435	183	2397	2580	0.622	7.564	8.186
0.095	44.8	15.40	188	2436	188	2394	2582	0.636	7.531	8.167
0.100	45.8	14.67	192	2437	192	2392	2584	0.649	7.500	8.149
0.12	49.4	12.36	207	2442	207	2383	2590	0.696	7.389	8.085
0.14	52.6	10.69	220	2446	220	2376	2596	0.737	7.294	8.031
0.16	55.3	9.432	232	2450	232	2369	2601	0.772	7.213	7.985
0.18	57.8	8.444	242	2453	242	2363	2605	0.804	7.140	7.944
0.20	60.1	7.648	251	2456	251	2358	2609	0.832	7.075	7.907
0.22	62.2	6.994	260	2459	260	2353	2613	0.858	7.016	7.874
0.24	64.1	6.445	268	2461	268	2348	2616	0.882	6.962	7.844
0.26	65.9	5.979	276	2464	276	2343	2619	0.904	6.913	7.817
0.28	67.5	5.578	283	2466	283	2339	2622	0.925	6.866	7.791
0.30	69.1	5.228	289	2468	289	2336	2625	0.944	6.823	7.767
0.32	70.6	4.921	295	2470	295	2332	2627	0.962	6.783	7.745
0.34	72.0	4.649	302	2472	302	2328	2630	0.980	6.745	7.725
0.36	73.4	4.407	307	2473	307	2325	2632	0.996	6.709	7.705
0.38	74.7	4.189	312	2475	312	2322	2634	1.011	6.675	7.686
0.40	75.9	3.992	318	2476	318	2318	2636	1.026	6.643	7.669
0.42	77.1	3.814	323	2478	323	2315	2638	1.040	6.612	7.652
0.44	78.2	3.651	327	2479	327	2313	2640	1.054	6.582	7.636
0.46	79.3	3.502	332	2481	332	2310	2642	1.067	6.554	7.621
0.48	80.3	3.366	336	2482	336	2308	2644	1.079	6.528	7.607
0.50	81.3	3.239	340	2483	340	2305	2645	1.091	6.502	7.593
0.55	83.7	2.964	351	2486	351	2298	2649	1.119	6.442	7.561
0.60	86.0	2.731	360	2489	360	2293	2653	1.145	6.386	7.531
0.65	88.0	2.535	369	2492	369	2288	2657	1.169	6.335	7.504
0.70	90.0	2.364	377	2494	377	2283	2660	1.192	6.286	7.478
0.75	91.8	2.217	384	2496	384	2278	2662	1.213	6.243	7.456
0.80	93.5	2.087	392	2498	392	2273	2665	1.233	6.201	7.434
0.85	95.2	1.972	399	2500	399	2269	2668	1.252	6.162	7.414
0.90	96.7	1.869	405	2502	405	2266	2671	1.270	6.124	7.394
0.95	98.2	1.777	411	2504	411	2262	2673	1.287	6.089	7.376
1.00	99.6	1.694	417	2506	417	2258	2675	1.303	6.056	7.359

$$\begin{aligned}
 * \quad \frac{h_f}{[\text{kJ/kg}]} &= \frac{pv_f}{[\text{kJ/kg}]} = \frac{p}{[\text{bar}]} \times \frac{10^5 [\text{N}]}{[\text{m}^2]} \times \frac{v_f}{[\text{m}^3/\text{kg}]} \times \left[\frac{\text{m}^3}{\text{kg}} \right] \times \frac{[\text{kJ}]}{10^3 [\text{N m}]} \times \frac{1}{[\text{kJ/kg}]} \\
 &= \frac{p}{[\text{bar}]} \times \frac{v_f}{[\text{m}^3/\text{kg}]} \times 10^2 = 0.006112 \times 0.0010002 \times 10^2 = 0.0006112
 \end{aligned}$$

p [bar]	T_s [°C]	v_g [m³/kg]	u_f [kJ/kg]	u_g [kJ/kg]	h_f [kJ/kg]	h_{fg} [kJ/kg]	h_g [kJ/kg]	s_f [kJ/kg K]	s_{fg} [kJ/kg K]	s_g [kJ/kg K]
1.0	99.6	1.694	417	2506	417	2258	2675	1.303	6.056	7.359
1.1	102.3	1.549	429	2510	429	2251	2680	1.333	5.994	7.327
1.2	104.8	1.428	439	2512	439	2244	2683	1.361	5.937	7.298
1.3	107.1	1.325	449	2515	449	2238	2687	1.387	5.884	7.271
1.4	109.3	1.236	458	2517	458	2232	2690	1.411	5.835	7.246
1.5	111.4	1.159	467	2519	467	2226	2693	1.434	5.789	7.223
1.6	113.3	1.091	475	2521	475	2221	2696	1.455	5.747	7.202
1.7	115.2	1.031	483	2524	483	2216	2699	1.475	5.707	7.182
1.8	116.9	0.9774	491	2526	491	2211	2702	1.494	5.669	7.163
1.9	118.6	0.9292	498	2528	498	2206	2704	1.513	5.632	7.145
2.0	120.2	0.8856	505	2530	505	2202	2707	1.530	5.597	7.127
2.1	121.8	0.8461	511	2531	511	2198	2709	1.547	5.564	7.111
2.2	123.3	0.8100	518	2533	518	2193	2711	1.563	5.533	7.096
2.3	124.7	0.7770	524	2534	524	2189	2713	1.578	5.503	7.081
2.4	126.1	0.7466	530	2536	530	2185	2715	1.593	5.474	7.067
2.5	127.4	0.7186	535	2537	535	2182	2717	1.607	5.446	7.053
2.6	128.7	0.6927	541	2539	541	2178	2719	1.621	5.419	7.040
2.7	130.0	0.6686	546	2540	546	2174	2720	1.634	5.393	7.027
2.8	131.2	0.6462	551	2541	551	2171	2722	1.647	5.368	7.015
2.9	132.4	0.6253	556	2543	556	2168	2724	1.660	5.344	7.004
3.0	133.5	0.6057	561	2544	561	2164	2725	1.672	5.321	6.993
3.5	138.9	0.5241	584	2549	584	2148	2732	1.727	5.214	6.941
4.0	143.6	0.4623	605	2554	605	2134	2739	1.776	5.121	6.897
4.5	147.9	0.4139	623	2558	623	2121	2744	1.820	5.037	6.857
5.0	151.8	0.3748	639	2562	640	2109	2749	1.860	4.962	6.822
5.5	155.5	0.3427	655	2565	656	2097	2753	1.897	4.893	6.790
6	158.8	0.3156	669	2568	670	2087	2757	1.931	4.830	6.761
7	165.0	0.2728	696	2573	697	2067	2764	1.992	4.717	6.709
8	170.4	0.2403	720	2577	721	2048	2769	2.046	4.617	6.663
9	175.4	0.2149	742	2581	743	2031	2774	2.094	4.529	6.623
10	179.9	0.1944	762	2584	763	2015	2778	2.138	4.448	6.586
11	184.1	0.1774	780	2586	781	2000	2781	2.179	4.375	6.554
12	188.0	0.1632	797	2588	798	1986	2784	2.216	4.307	6.523
13	191.6	0.1512	813	2590	815	1972	2787	2.251	4.244	6.495
14	195.0	0.1408	828	2593	830	1960	2790	2.284	4.185	6.469
15	198.3	0.1317	843	2595	845	1947	2792	2.315	4.130	6.445
16	201.4	0.1237	857	2596	859	1935	2794	2.344	4.078	6.422
17	204.3	0.1167	870	2597	872	1923	2795	2.372	4.028	6.400
18	207.1	0.1104	883	2598	885	1912	2797	2.398	3.981	6.379
19	209.8	0.1047	895	2599	897	1901	2798	2.423	3.936	6.359
20	212.4	0.09957	907	2600	909	1890	2799	2.447	3.893	6.340
22	217.2	0.09069	928	2601	931	1870	2801	2.492	3.813	6.305
24	221.8	0.08323	949	2602	952	1850	2802	2.534	3.738	6.272
26	226.0	0.07689	969	2603	972	1831	2803	2.574	3.668	6.242
28	230.0	0.07142	988	2603	991	1812	2803	2.611	3.602	6.213
30	233.8	0.06665	1004	2603	1008	1795	2803	2.645	3.541	6.186
32	237.4	0.06246	1021	2603	1025	1778	2803	2.679	3.482	6.161
34	240.9	0.05875	1038	2603	1042	1761	2803	2.710	3.426	6.136
36	244.2	0.05544	1054	2602	1058	1744	2802	2.740	3.373	6.113
38	247.3	0.05246	1068	2602	1073	1729	2802	2.769	3.322	6.091
40	250.3	0.04977	1082	2602	1087	1714	2801	2.797	3.273	6.070

p [bar]	T_s [°C]	v_g [m ³ /kg]	u_f u_g [kJ/kg]	h_f h_{fg} h_g [kJ/kg]	s_f s_{fg} s_g [kJ/kg K]
40	250.3	0.04977	1082 2602	1087 1714 2801	2.797 3.273 6.070
42	253.2	0.04732	1097 2601	1102 1698 2800	2.823 3.226 6.049
44	256.0	0.04509	1109 2600	1115 1683 2798	2.849 3.180 6.029
46	258.8	0.04305	1123 2599	1129 1668 2797	2.874 3.136 6.010
48	261.4	0.04117	1136 2598	1142 1654 2796	2.897 3.094 5.991
50	263.9	0.03944	1149 2597	1155 1639 2794	2.921 3.052 5.973
55	269.9	0.03563	1178 2594	1185 1605 2790	2.976 2.955 5.931
60	275.6	0.03244	1206 2590	1214 1570 2784	3.027 2.863 5.890
65	280.8	0.02972	1232 2586	1241 1538 2779	3.076 2.775 5.851
70	285.8	0.02737	1258 2581	1267 1505 2772	3.122 2.692 5.814
75	290.5	0.02532	1283 2576	1293 1473 2766	3.166 2.613 5.779
80	295.0	0.02352	1306 2570	1317 1441 2758	3.207 2.537 5.744
85	299.2	0.02192	1329 2565	1341 1410 2751	3.248 2.463 5.711
90	303.3	0.02048	1351 2559	1364 1379 2743	3.286 2.393 5.679
95	307.2	0.01919	1372 2552	1386 1348 2734	3.324 2.323 5.647
100	311.0	0.01802	1393 2545	1408 1317 2725	3.360 2.255 5.615
105	314.6	0.01696	1414 2537	1429 1286 2715	3.395 2.189 5.584
110	318.0	0.01598	1434 2529	1450 1255 2705	3.430 2.123 5.553
115	321.4	0.01508	1454 2522	1471 1224 2695	3.463 2.060 5.523
120	324.6	0.01426	1473 2514	1491 1194 2685	3.496 1.997 5.493
125	327.8	0.01349	1492 2505	1511 1163 2674	3.529 1.934 5.463
130	330.8	0.01278	1511 2496	1531 1131 2662	3.561 1.872 5.433
135	333.8	0.01211	1530 2487	1551 1099 2650	3.592 1.811 5.403
140	336.6	0.01149	1548 2477	1571 1067 2638	3.623 1.750 5.373
145	339.4	0.01090	1567 2467	1591 1034 2625	3.654 1.689 5.343
150	342.1	0.01035	1585 2456	1610 1001 2611	3.685 1.627 5.312
155	344.8	0.00982	1604 2445	1630 967 2597	3.715 1.565 5.280
160	347.3	0.00932	1623 2433	1650 932 2582	3.746 1.502 5.248
165	349.8	0.00884	1641 2420	1670 895 2565	3.777 1.437 5.214
170	352.3	0.00838	1660 2406	1690 858 2548	3.808 1.373 5.181
175	354.6	0.00794	1679 2391	1711 819 2530	3.839 1.305 5.144
180	357.0	0.00751	1699 2375	1732 778 2510	3.872 1.236 5.108
185	359.2	0.00709	1719 2358	1754 735 2489	3.905 1.163 5.068
190	361.4	0.00668	1740 2339	1777 689 2466	3.941 1.086 5.027
195	363.6	0.00627	1762 2318	1801 639 2440	3.977 1.004 4.981
200	365.7	0.00585	1786 2294	1827 584 2411	4.014 0.914 4.928
202	366.5	0.00569	1796 2283	1838 560 2398	4.031 0.875 4.906
204	367.4	0.00552	1806 2271	1849 535 2384	4.049 0.835 4.884
206	368.2	0.00534	1817 2259	1861 508 2369	4.067 0.792 4.859
208	369.0	0.00517	1829 2245	1874 479 2353	4.087 0.745 4.832
210	369.8	0.00498	1842 2231	1889 447 2336	4.108 0.695 4.803
212	370.6	0.00479	1856 2214	1904 412 2316	4.131 0.640 4.771
214	371.4	0.00458	1871 2196	1921 373 2294	4.157 0.579 4.736
216	372.1	0.00436	1888 2174	1940 328 2268	4.186 0.508 4.694
218	372.9	0.00409	1911 2146	1965 270 2235	4.224 0.417 4.641
220	373.7	0.00368	1949 2097	2008 170 2178	4.289 0.263 4.552
221.2	374.15	0.00317	2014 2014	2084 0 2084	4.406 0.000 4.406

$p/[\text{bar}]$ ($T_s/[^{\circ}\text{C}]$)		T [$^{\circ}\text{C}$]	50	100	150	200	250	300	400	500
0	$u = h - RT$ at $p = 0$	v								
		u	2446	2517	2589	2662	2737	2812	2969	3132
		h	2595	2689	2784	2880	2978	3077	3280	3489
		s								
0.006112 (0.01)	v_g 206.1	v	243.9	281.7	319.5	357.3	395.0	432.8	508.3	583.8
	u_g 2375	u	2446	2517	2589	2662	2737	2812	2969	3132
	h_g 2501	h	2595	2689	2784	2880	2978	3077	3280	3489
	s_g 9.155	s	9.468	9.739	9.978	10.193	10.390	10.571	10.897	11.187
0.01 (7.0)	v_g 129.2	v	149.1	172.2	195.3	218.4	241.4	264.5	310.7	356.8
	u_g 2385	u	2446	2517	2589	2662	2737	2812	2969	3132
	h_g 2514	h	2595	2689	2784	2880	2978	3077	3280	3489
	s_g 8.974	s	9.241	9.512	9.751	9.966	10.163	10.344	10.670	10.960
0.05 (32.9)	v_g 28.20	v	29.78	34.42	39.04	43.66	48.28	52.90	62.13	71.36
	u_g 2420	u	2445	2516	2589	2662	2737	2812	2969	3132
	h_g 2561	h	2594	2688	2784	2880	2978	3077	3280	3489
	s_g 8.394	s	8.496	8.768	9.008	9.223	9.420	9.601	9.927	10.217
0.1 (45.8)	v_g 14.67	v	14.87	17.20	19.51	21.83	24.14	26.45	31.06	35.68
	u_g 2437	u	2443	2516	2588	2662	2736	2812	2969	3132
	h_g 2584	h	2592	2688	2783	2880	2977	3077	3280	3489
	s_g 8.149	s	8.173	8.447	8.688	8.903	9.100	9.281	9.607	9.897
0.5 (81.3)	v_g 3.239	v		3.420	3.890	4.356	4.821	5.284	6.209	7.134
	u_g 2483	u		2512	2585	2660	2735	2812	2969	3132
	h_g 2645	h		2683	2780	2878	2976	3076	3279	3489
	s_g 7.593	s		7.694	7.940	8.158	8.355	8.537	8.864	9.154
0.75 (91.8)	v_g 2.217	v		2.271	2.588	2.901	3.211	3.521	4.138	4.755
	u_g 2496	u		2510	2585	2659	2734	2811	2969	3132
	h_g 2662	h		2680	2779	2877	2975	3075	3279	3489
	s_g 7.456	s		7.500	7.750	7.969	8.167	8.349	8.676	8.967
1 (99.6)	v_g 1.694	v		1.696	1.937	2.173	2.406	2.639	3.103	3.565
	u_g 2506	u		2506	2583	2659	2734	2811	2968	3131
	h_g 2675	h		2676	2777	2876	2975	3075	3278	3488
	s_g 7.359	s		7.360	7.614	7.834	8.033	8.215	8.543	8.834
1.01325 (100.0)	v_g 1.673	v			1.912	2.145	2.375	2.604	3.062	3.519
	u_g 2506	u			2583	2659	2734	2811	2968	3131
	h_g 2676	h			2777	2876	2975	3075	3278	3488
	s_g 7.355	s			7.608	7.828	8.027	8.209	8.537	8.828
1.5 (111.4)	v_g 1.159	v			1.286	1.445	1.601	1.757	2.067	2.376
	u_g 2519	u			2580	2656	2733	2809	2967	3131
	h_g 2693	h			2773	2873	2973	3073	3277	3488
	s_g 7.223	s			7.420	7.643	7.843	8.027	8.355	8.646
2 (120.2)	v_g 0.8856	v			0.9602	1.081	1.199	1.316	1.549	1.781
	u_g 2530	u			2578	2655	2731	2809	2967	3131
	h_g 2707	h			2770	2871	2971	3072	3277	3487
	s_g 7.127	s			7.280	7.507	7.708	7.892	8.221	8.513
3 (133.5)	v_g 0.6057	v			0.6342	0.7166	0.7965	0.8754	1.031	1.187
	u_g 2544	u			2572	2651	2729	2807	2966	3130
	h_g 2725	h			2762	2866	2968	3070	3275	3486
	s_g 6.993	s			7.078	7.312	7.517	7.702	8.032	8.324
4 (143.6)	v_g 0.4623	$r/[\text{m}^3/\text{kg}]$			0.4710	0.5345	0.5953	0.6549	0.7725	0.8893
	u_g 2554	$u/[\text{kJ}/\text{kg}]$			2565	2648	2727	2805	2965	3129
	h_g 2739	$h/[\text{kJ}/\text{kg}]$			2753	2862	2965	3067	3274	3485
	s_g 6.897	$s/[\text{kJ}/\text{kg K}]$			6.929	7.172	7.379	7.566	7.898	8.191

† The entries in all tables are regarded as pure numbers and therefore the symbols for the physical quantities should be divided by the appropriate units as shown for the entries at $p/[\text{bar}] = 4$. Because of lack of space, this has not been done consistently in the superheat and supercritical tables on pp. 6–9 and in the tables on

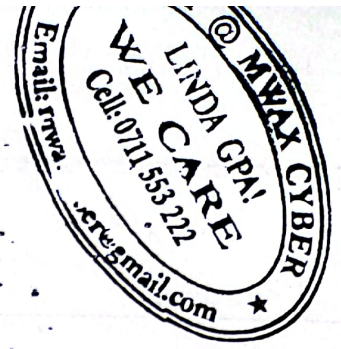
$p/[\text{bar}]$ ($T_s/[^{\circ}\text{C}]$)			T [$^{\circ}\text{C}$]	200	250	300	350	400	450	500	600
5 (151.8)	v_g	0.3748	v	0.4252	0.4745	0.5226	0.5701	0.6172	0.6641	0.7108	0.8040
	u_g	2562	u	2644	2725	2804	2883	2963	3045	3129	3300
	h_g	2749	h	2857	2962	3065	3168	3272	3377	3484	3702
	s_g	6.822	s	7.060	7.271	7.460	7.633	7.793	7.944	8.087	8.351
6 (158.8)	v_g	0.3156	v	0.3522	0.3940	0.4344	0.4743	0.5136	0.5528	0.5919	0.6697
	u_g	2568	u	2640	2722	2801	2881	2962	3044	3128	3299
	h_g	2757	h	2851	2958	3062	3166	3270	3376	3483	3701
	s_g	6.761	s	6.968	7.182	7.373	7.546	7.707	7.858	8.001	8.267
7 (165.0)	v_g	0.2728	v	0.3001	0.3364	0.3714	0.4058	0.4397	0.4734	0.5069	0.5737
	u_g	2573	u	2636	2720	2800	2880	2961	3043	3127	3298
	h_g	2764	h	2846	2955	3060	3164	3269	3374	3482	3700
	s_g	6.709	s	6.888	7.106	7.298	7.473	7.634	7.786	7.929	8.195
8 (170.4)	v_g	0.2403	v	0.2610	0.2933	0.3242	0.3544	0.3842	0.4138	0.4432	0.5018
	u_g	2577	u	2631	2716	2798	2878	2960	3042	3126	3298
	h_g	2769	h	2840	2951	3057	3162	3267	3373	3481	3699
	s_g	6.663	s	6.817	7.040	7.233	7.409	7.571	7.723	7.866	8.132
9 (175.4)	v_g	0.2149	v	0.2305	0.2597	0.2874	0.3144	0.3410	0.3674	0.3937	0.4458
	u_g	2581	u	2628	2714	2796	2877	2959	3041	3126	3298
	h_g	2774	h	2835	2948	3055	3160	3266	3372	3480	3699
	s_g	6.623	s	6.753	6.980	7.176	7.352	7.515	7.667	7.811	8.077
10 (179.9)	v_g	0.1944	v	0.2061	0.2328	0.2580	0.2825	0.3065	0.3303	0.3540	0.4010
	u_g	2584	u	2623	2711	2794	2875	2957	3040	3124	3297
	h_g	2778	h	2829	2944	3052	3158	3264	3370	3478	3698
	s_g	6.586	s	6.695	6.926	7.124	7.301	7.464	7.617	7.761	8.028
15 (198.3)	v_g	0.1317	v	0.1324	0.1520	0.1697	0.1865	0.2029	0.2191	0.2351	0.2667
	u_g	2595	u	2597	2697	2784	2868	2952	3035	3120	3294
	h_g	2792	h	2796	2925	3039	3148	3256	3364	3473	3694
	s_g	6.445	s	6.452	6.711	6.919	7.102	7.268	7.423	7.569	7.838
20 (212.4)	v_g	0.0996	v		0.1115	0.1255	0.1386	0.1511	0.1634	0.1756	0.1995
	u_g	2600	u		2681	2774	2861	2946	3030	3116	3291
	h_g	2799	h		2904	3025	3138	3248	3357	3467	3690
	s_g	6.340	s		6.547	6.768	6.957	7.126	7.283	7.431	7.701
30 (233.8)	v_g	0.0666	v		0.0706	0.0812	0.0905	0.0993	0.1078	0.1161	0.1324
	u_g	2603	u		2646	2751	2845	2933	3020	3108	3285
	h_g	2803	h		2858	2995	3117	3231	3343	3456	3682
	s_g	6.186	s		6.289	6.541	6.744	6.921	7.082	7.233	7.507
40 (250.3)	v_g	0.0498	v			0.0588	0.0664	0.0733	0.0800	0.0864	0.0988
	u_g	2602	u			2728	2828	2921	3010	3099	3279
	h_g	2801	h			2963	3094	3214	3330	3445	3674
	s_g	6.070	s			6.364	6.584	6.769	6.935	7.089	7.368
50 (263.9)	v_g	0.0394	v			0.0453	0.0519	0.0578	0.0632	0.0685	0.0786
	u_g	2597	u			2700	2810	2907	3000	3090	3273
	h_g	2794	h			2927	3070	3196	3316	3433	3666
	s_g	5.973	s			6.212	6.451	6.646	6.818	6.975	7.258
60 (275.6)	v_g	0.0324	v			0.0362	0.0422	0.0473	0.0521	0.0566	0.0652
	u_g	2590	u			2670	2792	2893	2988	3081	3266
	h_g	2784	h			2887	3045	3177	3301	3421	3657
	s_g	5.890	s			6.071	6.336	6.541	6.719	6.879	7.166
70 (285.8)	v_g	0.0274	$v/[\text{m}^3/\text{kg}]$			0.0295	0.0352	0.0399	0.0441	0.0481	0.0556
	u_g	2581	$u/[\text{kJ}/\text{kg}]$			2634	2772	2879	2978	3073	3260
	h_g	2772	$h/[\text{kJ}/\text{kg}]$			2841	3018	3158	3287	3410	3649
	s_g	5.814	$s/[\text{kJ}/\text{kg K}]$			5.934	6.231	6.448	6.632	6.796	7.088

* See footnote on p. 6.

$p/[\text{bar}]$ ($T_s/[^{\circ}\text{C}]$)		T [$^{\circ}\text{C}$]	350	375	400	425	450	500	600	700
80 (295.0)	v , 0.02352	$v/10^{-2}$	2.994	3.220	3.428	3.625	3.812	4.170	4.839	5.476
	h , 2758	h	2990	3067	3139	3207	3272	3398	3641	3881
	s , 5.744	s	6.133	6.255	6.364	6.463	6.555	6.723	7.019	7.279
90 (303.3)	v , 0.02048	$v/10^{-2}$	2.578	2.794	2.991	3.173	3.346	3.673	4.279	4.852
	h , 2743	h	2959	3042	3118	3189	3256	3385	3633	3874
	s_g 5.679	s	6.039	6.171	6.286	6.390	6.484	6.657	6.958	7.220
100 (311.0)	v , 0.01802	$v/10^{-2}$	2.241	2.453	2.639	2.812	2.972	3.275	3.831	4.353
	h , 2725	h	2926	3017	3097	3172	3241	3373	3624	3868
	s , 5.615	s	5.947	6.091	6.213	6.321	6.419	6.596	6.902	7.166
110 (318.0)	v , 0.01598	$v/10^{-2}$	1.960	2.169	2.350	2.514	2.666	2.949	3.465	3.945
	h , 2705	h	2889	2989	3075	3153	3225	3360	3616	3862
	s_g 5.553	s	5.856	6.014	6.143	6.257	6.358	6.539	6.850	7.117
120 (324.6)	v , 0.01426	$v/10^{-2}$	1.719	1.931	2.107	2.265	2.410	2.677	3.159	3.605
	h , 2685	h	2849	2960	3052	3134	3209	3348	3607	3856
	s_g 5.493	s	5.762	5.937	6.076	6.195	6.301	6.487	6.802	7.072
130 (330.8)	v_g 0.01278	$v/10^{-2}$	1.509	1.726	1.901	2.053	2.193	2.447	2.901	3.318
	h , 2662	h	2804	2929	3028	3114	3192	3335	3599	3850
	s , 5.433	s	5.664	5.862	6.011	6.136	6.246	6.437	6.758	7.030
140 (336.6)	v , 0.01149	$v/10^{-2}$	1.321	1.548	1.722	1.872	2.006	2.250	2.679	3.071
	h , 2638	h	2753	2896	3003	3093	3175	3322	3590	3843
	s_g 5.373	s	5.559	5.784	5.946	6.079	6.193	6.390	6.716	6.991
150 (342.1)	v , 0.01035	$v/10^{-2}$	1.146	1.391	1.566	1.714	1.844	2.078	2.487	2.857
	h , 2611	h	2693	2861	2977	3073	3157	3309	3581	3837
	s_g 5.312	s	5.443	5.707	5.883	6.023	6.142	6.345	6.677	6.954
160 (347.3)	v_g 0.00932	$v/10^{-2}$	0.976	1.248	1.427	1.573	1.702	1.928	2.319	2.670
	h , 2582	h	2617	2821	2949	3051	3139	3295	3573	3831
	s_g 5.248	s	5.304	5.626	5.820	5.968	6.093	6.301	6.639	6.919
170 (352.3)	v , 0.00838	$v/10^{-2}$		1.117	1.303	1.449	1.576	1.796	2.171	2.506
	h , 2548	h		2778	2920	3028	3121	3281	3564	3825
	s_g 5.181	s		5.541	5.756	5.914	6.044	6.260	6.603	6.886
180 (357.0)	v_g 0.00751	$v/10^{-2}$		0.997	1.191	1.338	1.463	1.678	2.039	2.359
	h , 2510	h		2729	2888	3004	3102	3268	3555	3818
	s_g 5.108	s		5.449	5.691	5.861	5.997	6.219	6.569	6.855
190 (361.4)	v , 0.00668	$v/10^{-2}$		0.882	1.089	1.238	1.362	1.572	1.921	2.228
	h , 2466	h		2674	2855	2980	3082	3254	3546	3812
	s_g 5.027	s		5.348	5.625	5.807	5.950	6.180	6.536	6.825
200 (365.7)	v , 0.00585	$v/10^{-2}[\text{m}^3/\text{kg}]$		0.768	0.995	1.147	1.270	1.477	1.815	2.110
	h , 2411	$h/[\text{kJ}/\text{kg}]$		2605	2819	2955	3062	3239	3537	3806
	s_g 4.928	$s/[\text{kJ}/\text{kg K}]$		5.228	5.556	5.753	5.904	6.142	6.505	6.796
210 (369.8)	v , 0.00498	$v/10^{-2}$		0.650	0.908	1.064	1.187	1.390	1.719	2.003
	h , 2336	h		2500	2781	2928	3041	3225	3528	3799
	s_g 4.803	s		5.050	5.484	5.699	5.859	6.105	6.474	6.768
220 (373.7)	v , 0.00368	$v/10^{-2}$		0.450	0.825	0.987	1.111	1.312	1.632	1.906
	h , 2178	h		2300	2738	2900	3020	3210	3519	3793
	s_g 4.552	s		4.725	5.409	5.645	5.813	6.068	6.444	6.742
221.2 (374.15)	v_c 0.00317	$v/10^{-2}$	0.163	0.351	0.816	0.978	1.103	1.303	1.622	1.895
	h_c 2084	h	1637	2139	2733	2896	3017	3208	3518	3792
	s_c 4.406	s	3.708	4.490	5.398	5.638	5.807	6.064	6.441	6.739

* See footnote on p. 6.

Note: linear interpolation is not accurate near the critical point.



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2019/2020

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN ELECTRICAL AND ELECTRONIC ENGINEERING

EEE 201: THERMODYNAMICS

DATE: Wednesday 29th January 2020

TIME: 11.00a.m -1.00p.m

INSTRUCTIONS:

This Paper consists of FIVE QUESTIONS

Answer Question ONE any Other TWO.

Steam Tables are provided

QUESTION ONE (COMPULSORY) 30 MARKS

- (a) Explain the following terms used in Engineering Thermodynamics citing types and examples where necessary. (3marks)
- (i) Open and Closed system
 - (ii) Process and Cycle
 - (iii) Intrinsic and Extrinsic Property
- (b) State the First Law of thermodynamics and Second law of Thermodynamics (3 marks)
- (c) State and explain Fourier's law of conduction and explain two mechanisms through which heat transfer by conduction in solids takes place (4Marks)
- (d) Calculate the rate of heat loss through the vertical walls of a boiler furnace of size 4m by 3 m by 3 m high. The walls are constructed from an inner fire brick wall 25 cm thick of thermal conductivity 0.4 W/mK, a layer of ceramic blanket insulation of thermal conductivity 0.2 W/mK and 8 cm thick, and a steel protective layer of thermal conductivity 55 W/mK and 2 mm thick. The inside temperature of the fire brick layer was measured at 600° C and the

temperature of the outside of the insulation 60°C . Also find the interface temperature of layers. (8 Marks)

- (e) One kg of a perfect gas is compressed from an initial state of 1 bar pressure and 25°C temperature to a final pressure of 6 bars. The polytropic index of compression, $n=1.25$. Considering the process as reversible, determine:

(i) Change in internal energy;

(ii) Compression work; $w = \frac{R(T_2 - T_1)}{n-1}$

(iii) Heat transferred by gas.

Given that molecular weight of gas = 30, Specific heat capacity of gas at constant pressure, $C_p = 1.7 \text{ kJ/kg K}$ and $R = 0.277 \text{ kJ/KgK}$ (6 Marks)

- f) Steam at a pressure of 2 bar and with dryness fraction 0.9 is compressed adiabatically to a pressure of 7 bar. The law of compression is $PV^{1.13} = C$. Determine:

i) The final dryness fraction of the steam

ii) The final density of the steam

iii) The change in internal energy of the steam

(6 Marks)

QUESTION 2 (20 MARKS)

- (a) Show that in a non flow polytropic expansion process, the heat transferred is given by

$$Q = \frac{\gamma - n}{\gamma - 1} \left(\frac{p_1 v_1 - p_2 v_2}{n - 1} \right)$$

Where Q = heat transferred

p_1, v_1 = initial pressure and volume respectively

p_2, v_2 = Final pressure and volume respectively

γ = adiabatic index

n = polytropic index of expansion/compression

(7marks)

- (b) Air at 1.02 bar, 22°C , initially occupying a cylinder volume of 0.015 m^3 , is compressed reversibly and adiabatically by a piston to a pressure of 6.8 bar. Calculate:

(i) The final temperature

(ii) The final volume

(iii) The work done on the mass of air in the cylinder.

(5 marks)

(c) A quantity of air occupies a volume of 30 litres at a temperature of 38°C and pressure 104 kN/m^2 . The temperature of the air can be raised;

- i) by isochoric heating process until the pressure is 208 kN/m^2
- ii) by adiabatic compression until volume is 6 litres.

Find for each case the final temperature, the external work done, the change in internal energy and the heat transferred. Take $R = 0.29 \text{ kJ/kg K}$ and $\gamma = 1.4$ (8 Marks)

QUESTION3 (20 MARKS)

(a) In an air compressor, air flows steadily at the rate of 0.2 kg/s . The condition of air at the inlet of compressor is given as: pressure = 0.9 bars ; specific volume = $0.9 \text{ m}^3/\text{kg}$; flow velocity = 5 m/s and temperature = 15°C . The condition of air at the discharge point is measured as 7.2 bars with specific volume of $0.2 \text{ m}^3/\text{kg}$, temperature 225°C and flow velocity 4 m/s . Heat taken away by cooling water from the compressor cylinder is 30 KJ/s . Calculate the power required to drive the compressor.

Calculate also the inlet and outlet pipe diameters.

Take specific heat at constant volume, C_v for air = 717 J/kg K (7 Marks)

(b) Show that for a perfect gas undergoing adiabatic expansion or compression,

$$\frac{T_1}{T_2} = \left(\frac{P_1}{P_2}\right)^{\frac{\gamma-1}{\gamma}} = \left(\frac{v_1}{v_2}\right)^{\gamma-1} \text{ where } T, P, v \text{ and } \gamma \text{ are Temperature, pressure, volume and adiabatic index respectively. (7 Marks)}$$

(c) Air flows steadily at a rate of 0.4 kg/s through an air compressor, entering at 6 m/s with a pressure of 1 bar and specific volume of $0.85 \text{ m}^3/\text{kg}$ and leaving at 4 m/s with a pressure of 6.9 bar and specific volume $0.16 \text{ m}^3/\text{kg}$. The specific internal energy of the air leaving is 88 kJ/kg greater than that of air entering. Cooling water jacket surrounding the cylinder absorbs heat from the air at rate of 59 kJ/s . Calculate the power required to drive the compressor and the inlet and outlet cross-sectional areas.

Assume that changes in potential energy is negligible. (6 Marks)

QUESTION 4 (20 MARKS)

(a) Show that the air standard thermal efficiency for a Joule cycle is given by

$$1 - \frac{1}{r^{\frac{\gamma-1}{\gamma}}} \quad (6 \text{ marks})$$

Where r = Compression ratio and γ = Adiabatic index

(b) In a standard OTTO cycle, the compression ratio is 7 and the compression begins at 1 bar and 313K. The heat added is 2510 kJ/kg. Find

(i) Maximum pressure and temperature of cycle;

(ii) Work done per kg of air

(iii) Cycle efficiency

(6 Marks)

(c) A gas turbine has an overall pressure ratio of 6/1 and a maximum cycle temperature of 600°C. The turbine drives the compressor and an electric generator. The ambient temperature is 20°C and the isentropic efficiencies of the compressor and turbine are 0.82 and 0.85 respectively. Calculate the power output in kilowatts for an air flow of 15kg/s. Calculate also the thermal efficiency and the work ratio. Neglect changes in kinetic energy, and the loss of pressure in the combustion chamber. Take for air $C_p = 1.005 \text{ kJ/kgK}$ and $\gamma = 1.4$. For combustion gases $C_p = 1.15 \text{ kJ/kgK}$ and $\gamma = 1.33$

(8 Marks)

QUESTIONS 5 (20 MARKS)

(a) Define dryness fraction of steam

(1 Mark)

(b) Using a temperature-enthalpy diagram, show the different phases of steam generation indicating type of enthalpy at each phase.

(4 Marks)

(c) 1 Kg of 0.5 dry steam at 6.0 bar pressure is heated, so that it becomes

(i) 0.95 dry at 6.0 bar pressure ✓

(ii) Dry & saturated at 6.0 bar

(iii) Superheated to 300°C at 6.0 bar

(iv) Superheated to 250°C at 8.0 bar

(6 Marks)

Using steam tables determine in each case the quantity of heat required to be supplied. Take C_{sup} for superheated steam as 2.3 kJ/ kg K.

(d) Steam is supplied; dry saturated at 40 bar to a turbine and the condenser pressure is 0.035 bar. If the plant operates on the Rankine cycle, calculate, per kg of steam:

i. The work output neglecting the feed pump work;

ii. The heat supplied;

iii. The specific steam consumption

iv. Thermal Efficiency of the plant

(9 Marks)

Now that in a non flow polytropic process, the ratio of the heat transferred to the work done is given by

$$Q = W \frac{\gamma - n}{\gamma - 1}$$

Where Q = heat transferred, W = the work done, γ = adiabatic index
 n = polytropic index of expansion/compression

1. Air flows steadily at a rate of 0.4 kg/s through an air compressor, entering at 6 m/s with a pressure of 1 bar and specific volume of 0.85 m³/kg and leaving at 4 m/s with a pressure of 6.9 bar and specific volume 0.16 m³/kg. The specific internal energy of the air leaving is 88 kJ/kg greater than that of air entering. Cooling water jacket surrounding the cylinder absorbs heat from the air at rate of 59 kJ/s. Calculate the power required to drive the compressor and the inlet and outlet cross-sectional areas. Assume that changes in potential energy is negligible.
2. Air at 1.02 bar, 22°C, initially occupying a cylinder volume of 0.015 m³, is compressed reversibly and adiabatically by a piston to a pressure of 6.8 bar. Calculate:
- The final temperature
 - The final volume
 - The work done on the mass of air in the cylinder.





KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2016/2017

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE (ELECTRICAL AND ELECTRONIC ENGINEERING) AND BACHELOR OF
SCIENCE (BIOMEDICAL ENGINEERING)

EEE 201: THERMODYNAMICS

DATE: Tuesday 6th December 2016

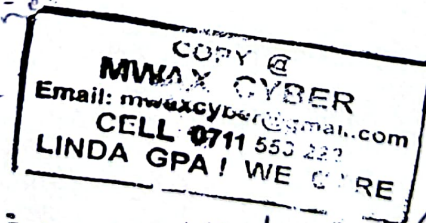
TIME: 2.00p.m-4.00p.m

INSTRUCTIONS:

- This paper contains FIVE questions.
- Question 1 is compulsory and then respond to TWO other questions.

Question 1 (30 Marks)

- a) What is thermodynamics? (2 Marks)
- b) Why should we study thermodynamics? (2 Marks)
- c) Define the following terms as they relate to thermodynamics. (11 Marks)
- (1 each)
- i. System ✓
 - ii. Boundary
 - iii. Surroundings
 - iv. Closed System
 - v. Open System
 - vi. Simple Fluid ✓ *has single element*
 - vii. Non-flow processes
 - viii. Flow Processes
 - ix. Equilibrium
 - x. State *- condition of a system at any particular time (set of properties)*
 - xi. Properties
- d) How many properties are necessary and sufficient to determine the state of a closed system containing a simple fluid? (1 Mark)



Volume
Pressure
Temp

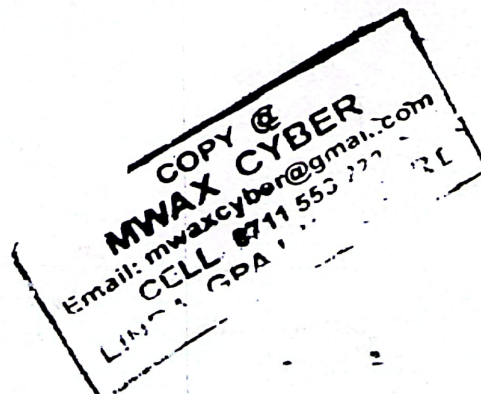
Pressure, Volume

- e) What mechanical properties are necessary to determine mechanical state during the analysis of an open system? (2 Marks)
- f) State the First and Second Laws of Thermodynamics. (2 Marks)
- g) Differentiate between a reversible and irreversible thermodynamic process. (4 Marks)
- h) State the Zeroth law of thermodynamics. (2 Marks)
- i) Define work and heat in the context of thermodynamics. (4 Marks)

Energy

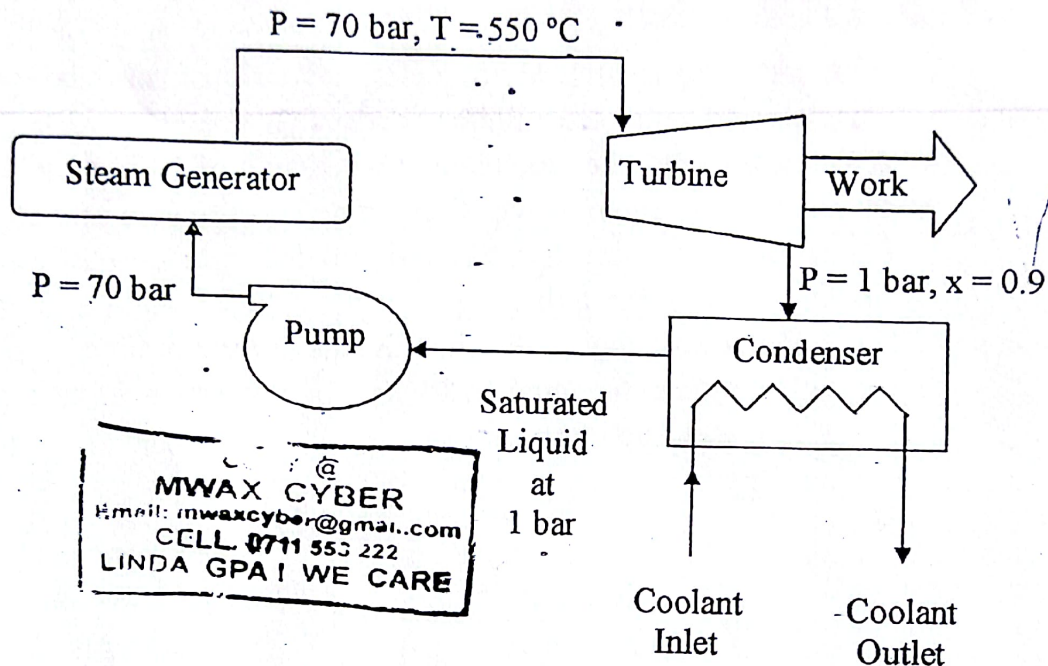
Question 2 (20 Marks)

- a) What are the three modes of heat transfer? (3 Marks)
- b) Describe how heat energy flows in each of the above modes and give a practical example of each. (9 Marks)
- * c) A wall made of material A, 250 mm thick, is faced with material B, 50 mm thick. Material A has a coefficient of thermal conductivity of $0.69 \text{ W/m}\cdot\text{K}$ while that of material B is $0.93 \text{ W/m}\cdot\text{K}$. If the temperature of the exposed material A face is 30°C and that of material B is 5°C .
 - i. Determine the heat lost in 1 hour through a wall 10 meter long by 5 meters high. (4 Marks)
 - ii. Determine the interface temperature. (4 Marks)



Question 3 (20 Marks)

For the simple steam power plant shown below, answer the following questions given that the fluid flow rate is 40 kg/sec:



- What is the power output of the turbine? (7 Marks)
- What is the heat transfer in the condenser? (3 Marks)
- What is the heat transfer in the steam generator? (3 Marks)
- Water to serve as a coolant in the condenser is available from a nearby river at a temperature of 20°C and exits the condenser at 30°C. How many kilograms per second of this water are required? (4 Marks)
- What is the Carnot efficiency of the process above if the heat source for the steam generator is a flue gas at 650°C? (3 Marks)

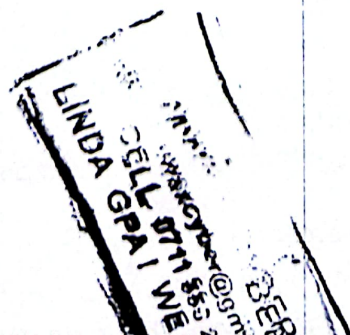
Question 4 (20 marks)

- Write the non-flow equation for a constant volume process. (1 Mark)
- Write the non-flow equation for a constant pressure process. (1 Mark)
- Differentiate between internal energy and enthalpy. (2 Marks)
- Define the terms adiabatic and isothermal. (2 Marks)
- A constant volume non-flow process containing 1.5 kg of water as the working fluid transitions from state 1 (Superheated Vapour, 500 °C, 3 bar) to state 2 (Saturated Vapour at 2 bar, $x=0.6$).
 - What is the heat transfer for this process and in what direction is it? (8 Marks)
 - If the heat was taken up by a cooling coil in 3 minutes, what is the power rating of this cooling coil? (2 Marks)

iii. What is the entropy change for this process? (4 marks)

Question 5 (20 Marks)

- a) With regard to isentropic processes, beyond just zero change in entropy; what two attributes are unique about such processes? (1 Mark)
- b) A steam turbine receives steam at a pressure of 10 bar and a temperature of 300 °C. The steam leaves the turbine at a pressure of 0.15 bar. The work output of the turbine is measured and found to be 600 KJ/kg of steam flowing through the turbine.
- Determine the isentropic efficiency of the turbine. (15 Mark)
 - Determine the quality/dryness of the steam at exit point for the non-isentropic process. (4 Marks)



- Define the following
- Electrical Power (1 mark)
 - Electric Energy (1 mark)
 - Conductance (1 mark)
 - State
 - Thevenin's theorem (3 marks)
 - Superposition theorem (3 marks)
- d) In the network shown Fig 2(c) find the different branch currents by superposition theorem. (5 marks)

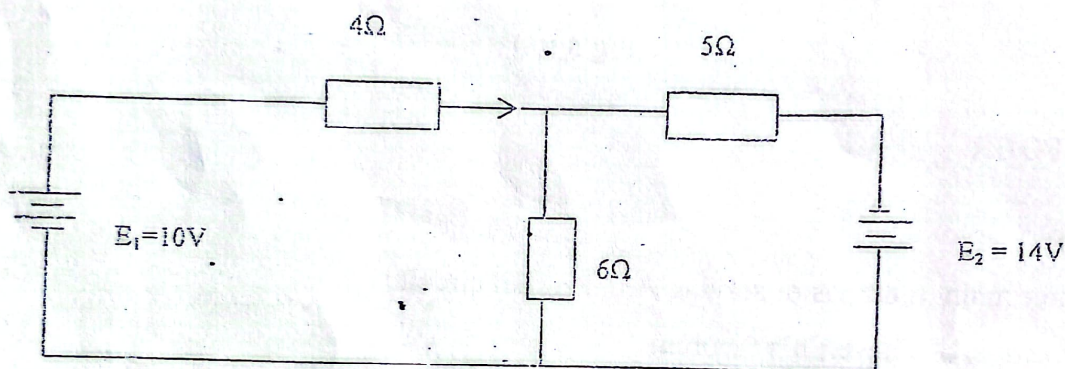


Fig 2 (c)

QUESTION THREE

- a) Define these terms used in circuit analysis
- Resistance (R) (2 marks)
 - Capacitance (C) (2 marks)
 - Inductance (L) (2 marks)

Show how they relate to each other (3 marks)

- b) A coil of p.f 0.8 is connected in series with a $110\mu\text{F}$ Capacitor. The frequency is 50Hz. The p.d across the coil is found to be equal to the p.d across the capacitor, Calculate the resistance and inductance of the coil (5 marks)
- c) Using Norton's theorem, find the current flowing in the branch AB containing 6Ω of the network in the Figure 4 (c) (6 marks)

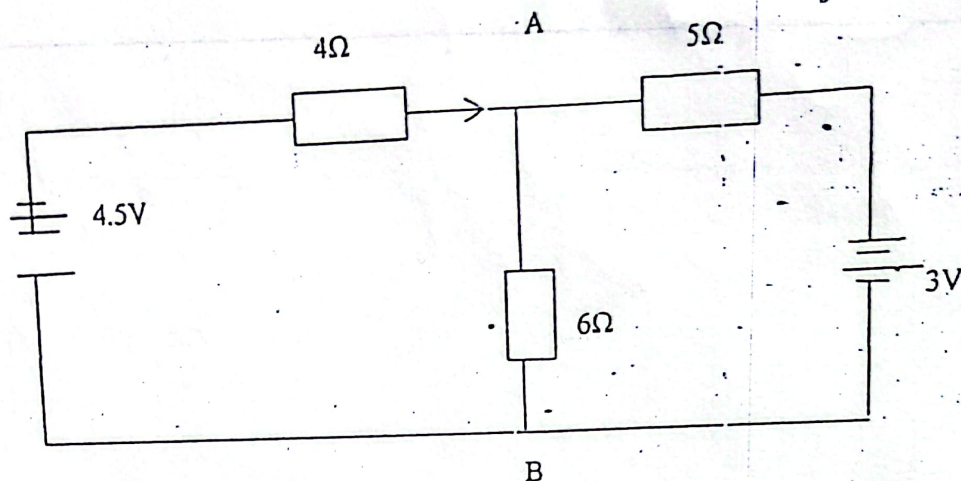


Fig 3 (c)

QUESTION FOUR

- a) Explain
 - i) The three main methods of solving A.C Parallel circuits (3 marks)
 - ii) The converse of Thevenin's theorem (3 marks)
 - iii) When is it appropriate to use Star/Delta Transformation (2 marks)
- a) A voltage of $230 \angle 30^\circ$ V is applied to two circuits connected in parallel. The current in the branches are $20 \angle 60^\circ$ A and $40 \angle -30^\circ$ A.
Find
 - i) The total impedance of the circuit (3 marks)
 - ii) Active and apparent power of each branch (2 marks)
- b) Two impedances $Z_1 = 2 + j3$ and $Z_2 = 3 - j4$ are connected in parallel across 250V, 50Hz supply.
Calculate
 - i) The current in each branch (2 marks)
 - ii) The line current (2 marks)
 - iii) Circuit p.f (1 mark)
 - iv) Power taken by the circuit (2 marks)

a) Define the following terminologies

- | | |
|------------------|-----------|
| i) Impedance | (2 marks) |
| ii) Admittance | (1 mark) |
| iii) Conductance | (1 mark) |
| iv) Susceptance | (1 mark) |

Show how they relate to each other and indicate their units (2 marks)

b) A 230V, 50Hz a.c supply is applied to a coil of 0.06H Inductance and 2.5Ω resistance connected in series with a $6.8\mu\text{F}$ capacitor.

Calculate

- | | |
|--|-----------|
| i) Impedance | (2 marks) |
| ii) Current | (2 marks) |
| iii) Phase angle between current and voltage | (1 mark) |
| iv) Power factor | (1 mark) |
| v) Power consumed | (1 mark) |
- c) A single phase motor operating off a 400V, 50Hz supply is developing 10kw with an efficiency of 84% and Power factor of 0.7 lagging
- d) Calculate
- e) A) The input apparent power (2 marks)
- f) The active and reactive component of current (2 marks)
- g) The reactive power (in kilovars) (2 marks)



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2019/2020

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE IN ELECTRICAL AND ELECTRONIC ENGINEERING

EEE 201: THERMODYNAMICS

DATE: Wednesday 29th January 2020

TIME: 11.00a.m -1.00p.m

INSTRUCTIONS:

This Paper consists of FIVE QUESTIONS

Answer Question ONE any Other TWO.

Steam Tables are provided

QUESTION ONE (COMPULSORY) 30 MARKS

(a) Explain the following terms used in Engineering Thermodynamics citing types and examples where necessary.

(i) Open and Closed system

(ii) Process and Cycle

(iii) Intrinsic and Extrinsic Property

(3marks)

(b) State the First Law of thermodynamics and Second law of Thermodynamics

(3 marks)

(c) State and explain Fourier's law of conduction and explain two mechanisms through which heat transfer by conduction in solids takes place

(4Marks)

(d) Calculate the rate of heat loss through the vertical walls of a boiler furnace of size 4m by 3 m by 3 m high. The walls are constructed from an inner fire brick wall 25 cm thick of thermal conductivity 0.4 W/mK, a layer of ceramic blanket insulation of thermal conductivity 0.2 W/mK and 8 cm thick, and a steel protective layer of thermal conductivity 55 W/mK and 2 mm thick. The inside temperature of the fire brick layer was measured at 600° C and the

temperature of the outside of the insulation 60°C . Also find the interface temperature of layers. (8 Marks)

- (e) One kg of a perfect gas is compressed from an initial state of 1 bar pressure and 25°C temperature to a final pressure of 6 bars. The polytropic index of compression, $n=1.25$. Considering the process as reversible, determine:

- (i) Change in internal energy;
- (ii) Compression work;
- (iii) Heat transferred by gas.

Given that molecular weight of gas = 30; Specific heat capacity of gas at constant pressure, $C_p = 1.7 \text{ kJ/kg K}$ and $R = 0.277 \text{ kJ/KgK}$ (6 Marks)

- f) Steam at a pressure of 2 bar and with dryness fraction 0.9 is compressed adiabatically to a pressure of 7 bar. The law of compression is $PV^{1.13} = C$. Determine:
- i) The final dryness fraction of the steam
 - ii) The final density of the steam
 - iii) The change in internal energy of the steam
- (6 Marks)

QUESTION 2 (20 MARKS)

- (a) Show that in a non flow polytropic expansion process, the heat transferred is given by

$$Q = \frac{\gamma - n}{\gamma - 1} \left(\frac{p_1 v_1 - p_2 v_2}{n - 1} \right)$$

Where Q = heat transferred

p_1, v_1 = initial pressure and volume respectively

p_2, v_2 = Final pressure and volume respectively

γ = adiabatic index

n = polytropic index of expansion/compression (7marks)

- (b) Air at 1.02 bar, 22°C , initially occupying a cylinder volume of 0.015 m^3 , is compressed reversibly and adiabatically by a piston to a pressure of 6.8 bar. Calculate:

- (i) The final temperature
 - (ii) The final volume
 - (iii) The work done on the mass of air in the cylinder.
- (5 marks)

(c) A quantity of air occupies a volume of 30 litres at a temperature of 38 °C and pressure 104 kN/m². The temperature of the air can be raised;

- i) by isochoric heating process until the pressure is 208 kN/m²
- ii) by adiabatic compression until volume is 6 litres.

Find for each case the final temperature, the external work done, the change in internal energy and the heat transferred. Take $R = 0.29 \text{ kJ/kg K}$ and $\gamma = 1.4$ - (8 Marks)

QUESTION3 (20 MARKS)

(a) In an air compressor, air flows steadily at the rate of 0.2 kg/s. The condition of air at the inlet of compressor is given as: pressure = 0.9 bars; specific volume = 0.9 m³/kg; flow velocity = 5 m/s and temperature = 15°C. The condition of air at the discharge point is measured as 7.2 bars with specific volume of 0.2 m³/kg, temperature 225°C and flow velocity 4m/s. Heat taken away by cooling water from the compressor cylinder is 30 KJ/s. Calculate the power required to drive the compressor.

Calculate also the inlet and outlet pipe diameters.

Take specific heat at constant volume, C_v for air = 717 J/kg K

(7 Marks)

(b) Show that for a perfect gas undergoing adiabatic expansion or compression,

$$\frac{T_1}{T_2} = \left(\frac{P_1}{P_2}\right)^{\frac{\gamma-1}{\gamma}} = \left(\frac{v_1}{v_2}\right)^{\gamma-1} \text{ where } T, P, v \text{ and } \gamma \text{ are Temperature, pressure, volume and}$$

adiabatic index respectively.

(7 Marks)

(c) Air flows steadily at a rate of 0.4kg/s through an air compressor, entering at 6m/s with a pressure of 1 bar and specific volume of 0.85m³/kg and leaving at 4m/s with a pressure of 6.9bar and specific volume 0.16m³/kg. The specific internal energy of the air leaving is 88kJ/kg greater than that of air entering. Cooling water jacket surrounding the cylinder absorbs heat from the air at rate of 59kJ/s. Calculate the power required to drive the compressor and the inlet and outlet cross-sectional areas.

Assume that changes in potential energy is negligible.

(6 Marks)

QUESTION 4 (20 MARKS)

(a) Show that the air standard thermal efficiency for a Joule cycle is given by

$$1 - \frac{1}{r^{\frac{\gamma-1}{\gamma}}}$$

(6 marks)

Where r = Compression ratio and γ = Adiabatic index

(b) In a standard OTTO cycle, the compression ratio is 7 and the compression begins at 1 bar and 313K. The heat added is 2510 kJ/kg. Find

(i) Maximum pressure and temperature of cycle;

(ii) Work done per kg of air

(iii) Cycle efficiency

(6 Marks)

(c) A gas turbine has an overall pressure ratio of 6/1 and a maximum cycle temperature of 600°C . The turbine drives the compressor and an electric generator. The ambient temperature is 20°C and the isentropic efficiencies of the compressor and turbine are 0.82 and 0.85 respectively. Calculate the power output in kilowatts for an air flow of 15kg/s. Calculate also the thermal efficiency and the work ratio. Neglect changes in kinetic energy, and the loss of pressure in the combustion chamber. Take for air $C_p = 1.005 \text{ kJ/kgK}$ and $\gamma = 1.4$. For combustion gases $C_p = 1.15 \text{ kJ/kgK}$ and $\gamma = 1.33$

(8 Marks)

QUESTIONS 5 (20 MARKS)

(a) Define dryness fraction of steam

(1 Mark)

(b) Using a temperature-enthalpy diagram, show the different phases of steam generation indicating type of enthalpy at each phase.

(4 Marks)

(c) 1 Kg of 0.5 dry steam at 6.0 bar pressure is heated, so that it becomes

(i) 0.95 dry at 6.0 bar pressure

(ii) Dry & saturated at 6.0 bar

(iii) Superheated to 300°C at 6.0 bar

(iv) Superheated to 250°C at 8.0 bar

(6 Marks)

Using steam tables determine in each case the quantity of heat required to be supplied. Take C_{sup} for superheated steam as 2.3 kJ/kg K .

(d) Steam is supplied; dry saturated at 40 bar to a turbine and the condenser pressure is 0.035 bar. If the plant operates on the Rankine cycle, calculate, per kg of steam:

i. The work output neglecting the feed pump work;

ii. The heat supplied;

iii. The specific steam consumption

iv. Thermal Efficiency of the plant

(9 Marks)



STUDENT CENTRE K.M.
Email: studentcentre@yaho.com

KENYATTA UNIVERSITY

UNIVERSITY EXAMINATION 2012/2013 ACADEMIC YEAR

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
ENGINEERING (ELECTRICAL AND ELECTRONICS)

EEE 201: Thermodynamics

DATE: 19/10/2012

TIME: 9:00 TO 10:00am

Question 1.

a) Define the following terms

- i) Work ✓
- ii) Heat ✓
- iii) Internal energy ✓
- iv) Enthalpy ✓

(8 Marks)

b) Explain the following

- i) First Law of Thermodynamic ✓
- ii) Steady flow energy equation ✓
- iii) Second law of Thermodynamics ✓
- iv) Entropy ✓

(8 Marks)

c) In the turbine of a gas turbine unit the gas flow through the turbine at 18 kg/s and the power developed by the turbine is 16000 KW . The specific enthalpy of the gases at inlet and outlet are 1100 KJ/Kg and 380 KJ/Kg respectively and the velocity of the gases at inlet and outlet are 60 m/s and 150 m/s respectively

- i) The rate at which heat is rejected from the turbine
- ii) Area of the inlet pipe given that the specific volume of the gases at inlet is $0.5 \text{ m}^3/\text{kg}$

(14 Marks)

$$\frac{dQ}{dt} = \frac{dQ}{dt} + \frac{dW}{dt} = \frac{dQ}{dt} + \frac{dW}{dt}$$

$$F \cdot Q \cdot Q = 0$$

$$\frac{1}{A} = \frac{V}{Q}$$

$$\frac{18 \text{ kg/s} \times 0.5 \text{ m}^3/\text{kg}}{18 \text{ kg/s} \times 0.5 \text{ m}^3/\text{kg}}$$

$$18 \text{ kg/s} \times 0.5 \text{ m}^3/\text{kg}$$